

SIMATS ENGINEERING

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EXPERIMENT 6

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MACHINE LEARNING

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import CategoricalNB
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
from sklearn.metrics import classification_report

df = pd.read_csv("/content/play_tennis (1).csv")

print("===== DATASET =====\n")
display(df)

print("\nDataset Shape:", df.shape)

print("\n===== MISSING VALUES =====\n")
print(df.isnull().sum())

df = df.dropna()

X = df.iloc[:, :-1]
y = df.iloc[:, -1]

print("\n===== INPUT FEATURES =====\n")
display(X)
```

```
print("\n===== TARGET =====\n")
```

```
print(y)
```

```
label_encoders = {}
```

```
for column in X.columns:
```

```
    encoder = LabelEncoder()
```

```
    X[column] = encoder.fit_transform(X[column])
```

```
    label_encoders[column] = encoder
```

```
print("\n===== ENCODED FEATURES =====\n")
```

```
display(X)
```

```
target_encoder = LabelEncoder()
```

```
y = target_encoder.fit_transform(y)
```

```
print("\n===== ENCODED TARGET =====\n")
```

```
print(y)
```

```
print("\nTarget Encoding:")
```

```
for i, label in enumerate(target_encoder.classes_):
```

```
    print(label, "=", i)
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X,
```

```
    y,
```

```
    test_size=0.2,
```

```
    random_state=42
```

)

```
print("\n===== DATA SPLIT =====\n")
```

```
print("Training samples:", len(X_train))
```

```
print("Testing samples :", len(X_test))
```

```
model = CategoricalNB()
```

```
model.fit(X_train, y_train)
```

```
print("\n===== TRAINING =====\n")
```

```
print("Naive Bayes training completed successfully!")
```

```
y_pred = model.predict(X_test)
```

```
print("\n===== PREDICTION =====\n")
```

```
print("Actual values  :", y_test)
```

```
print("Predicted values :", y_pred)
```

```
# Convert encoded values back to Yes / No
```

```
actual = target_encoder.inverse_transform(y_test)
```

```
predicted = target_encoder.inverse_transform(y_pred)
```

```
print("\nActual Classes  :", actual)
```

```
print("Predicted Classes :", predicted)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\n===== ACCURACY =====\n")
```

```
print("Accuracy:", accuracy)
```

```
print("Accuracy (%):", accuracy * 100)
```

```
print("\n===== CLASSIFICATION REPORT =====\n")
```

```
print(
    classification_report(
        y_test,
        y_pred,
        target_names=target_encoder.classes_,
        zero_division=0
    )
)
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("\n===== CONFUSION MATRIX =====\n")
```

```
print(cm)
```

```
# Display confusion matrix
```

```
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=target_encoder.classes_
)
```

```
disp.plot()
```

```
plt.title("Naive Bayes Confusion Matrix")
```

```
plt.show()
```

```
print("\n=====")
```

```
print("FINAL RESULT")
```

```
print("=====")
```

```
print("Algorithm : Naive Bayes")
```

```
print("Dataset : Play Tennis")
```

```
print("Accuracy :", round(accuracy * 100, 2), "%")
```

```
print("=====")
```

OUTPUT:

===== INPUT FEATURES =====

	day	outlook	temp	humidity	wind	
0	D1	Sunny	Hot	High	Weak	
1	D2	Sunny	Hot	High	Strong	
2	D3	Overcast	Hot	High	Weak	
3	D4	Rain	Mild	High	Weak	
4	D5	Rain	Cool	Normal	Weak	
5	D6	Rain	Cool	Normal	Strong	
6	D7	Overcast	Cool	Normal	Strong	
7	D8	Sunny	Mild	High	Weak	
8	D9	Sunny	Cool	Normal	Weak	
9	D10	Rain	Mild	Normal	Weak	
10	D11	Sunny	Mild	Normal	Strong	
11	D12	Overcast	Mild	High	Strong	
12	D13	Overcast	Hot	Normal	Weak	
13	D14	Rain	Mild	High	Strong	

	day	outlook	temp	humidity	wind	
0	0	2	1	0	1	
1	6	2	1	0	0	
2	7	0	1	0	1	
3	8	1	2	0	1	
4	9	1	0	1	1	
5	10	1	0	1	0	
6	11	0	0	1	0	
7	12	2	2	0	1	
8	13	2	0	1	1	
9	1	1	2	1	1	
10	2	2	2	1	0	
11	3	0	2	0	0	
12	4	0	1	1	1	
13	5	1	2	0	0	



```

... ===== ENCODED TARGET =====

[0 0 1 1 1 0 1 0 1 1 1 1 1 0]

Target Encoding:
No = 0
Yes = 1

===== DATA SPLIT =====

Training samples: 11
Testing samples : 3

===== TRAINING =====

Naive Bayes training completed successfully!

===== PREDICTION =====

Actual values      : [1 1 0]
Predicted values   : [1 0 1]

Actual Classes     : ['Yes' 'Yes' 'No']
Predicted Classes  : ['Yes' 'No' 'Yes']

===== ACCURACY =====

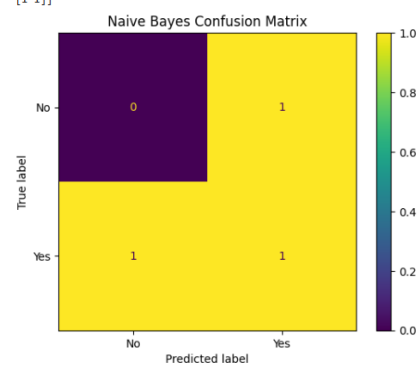
Accuracy: 0.3333333333333333
Accuracy (%): 33.33333333333333

```

```
*** ===== CLASSIFICATION REPORT =====  
  
              precision    recall  f1-score   support  
  
   No           0.00        0.00        0.00         1  
   Yes          0.50        0.50        0.50         2  
  
 accuracy              0.33         3  
 macro avg           0.25        0.25        0.25         3  
 weighted avg        0.33        0.33        0.33         3
```

===== CONFUSION MATRIX =====

```
[[0 1]  
 [1 1]]
```



=====

FINAL RESULT

=====

Algorithm : Naive Bayes

Dataset : Play Tennis

Accuracy : 33.33 %

=====
